STATS 503, Homework 6

Please use a Jupyter notebook to write up your solutions. Submit your work through Canvas by uploading an html file that contains the html export of your notebook file. The html file should include all your code (and therefore no hidden cells). Use markdown formatting to indicate which part of your notebook corresponds to each problem.

1. [**5 POINTS**] ISLP chapter 7 exercise #1
2. [**5 POINTS**] ISLP chapter 7 exercise #2. For this problem, there are two important points to consider:
   - Each part of the problem requires you to create a "sketch." By sketch, we mean an actual drawing. This could be a photograph of something you have hand-drawn on paper or an image created using a digital drawing tool. While creating a plot with matplotlib is also an option, it might require more effort than necessary for this task. The sketches don't need to be perfect, but should include a vertical axis, a horizontal axis, at least six points that represent plausible training data, and a basic illustration of the curve $\hat{g}$.
   - Regarding part (e) of the problem, which looks at the scenario when $\lambda = 0$ and $m = 3$. In this scenario, the curve $\hat{g}$ is not uniquely defined; there are many different answers that will all be considered correct. However, this doesn't mean that any sketch will be considered correct.
3. [**5 POINTS**] ISLP chapter 7 exercise #3. This requires another sketch. As before, a drawing is what we're looking for. However, for this sketch, as per the instructions in the textbook problem, it must include annotations detailing "intercepts, slopes, and other relevant information." More specifically, the sketch should clearly show:
   - The horizontal and vertical axes of the plot.
   - The point where the curve crosses the y-axis.
   - The point(s) where the curve intersects the x-axis, if any, within the range of x values being sketched.
   - The slope of any straight segments of the curve.
4. [**5 POINTS**] ISLP chapter 7 exercise #4 (cf. remarks for the previous problem)
5. [**2 POINTS**] Which of the following statements about local regression (e.g., LOESS) is true?
   (a) It is used primarily to predict categorical outcomes.
   (b) It is less flexible compared to global regression models.
   (c) It tends to underfit when the span is large.

(d) It assumes that the relationship between the predictors and the responses is linear throughout the domain of the predictors.

6. [**2 POINTS**] What is the primary difference between cubic splines and natural cubic splines?

   (a) Cubic splines are defined for each interval between data points, while natural cubic splines are only defined at the data points themselves.

   (b) Cubic splines require the second derivative at the endpoints to be zero, whereas natural cubic splines do not have any restrictions on the derivatives.

   (c) Natural cubic splines ensure that the second derivative at the endpoints of the spline is zero, while cubic splines do not have this restriction.

   (d) Cubic splines are smoother and have higher continuity than natural cubic splines, which tend to have sharp turns at the data points.

7. This question uses the following variables from the `boston` data, which can be downloaded from canvas. We are interested in predicting the air quality variable `nox`.

   (a) [**3 POINTS**] For this problem, we will estimate the conditional expectation of `nox` given `indus` (one predictor). We will consider three different estimators:

      i. Linear regression with spline features and knots at $\{0, 12, 17, 30\}$
      ii. Linear regression with polynomial features of degree 12
      iii. K-nearest-neighbors regression with 13 neighbors

      Estimate the mean squared predictive error for each estimator using 4-fold cross-validation (remember to use the same splits for each estimator). Report the error for each of the estimators. Which estimator gives the lowest error?

      Fit each estimator to the entire dataset and plot the three predicted curves against a scatterplot of `indus` vs `nox` in the data. Set `alpha=0.1` in the scatterplot. (Otherwise, you will not be able to tell that there are many points that are "on top of each other"). Why might one choose to highlight the curve with the *highest* error when discussing air quality in a public presentation? A short explanation of one to three sentences will suffice.

   (b) [**3 POINTS**] Use `pygam` to predict `nox` from `dis`, `indus`, and `rad`. Treat each of these variables as continuous.

      i. Use `pygam` to fit the data, considering different choices for the `lam` that controls the regularization strength. For each feature, consider four choices of regularization, $\lambda \in \{0.5, 1, 5, 10\}$. (There will be $4^3 = 64$ fits in total.) Use `pygam`'s gridsearch method to identify the best choice. Report the best λ value found.
      ii. Plot the partial dependence for each of the three features.

iii. The feature `rad` indicates an index of accessibility to highways. How well do you think the estimated partial dependence for this feature represents the true relationship between `rad` and `nox`? Explain.

Total points for this assignment: 30